

QBITEL BRIDGE

Automotive & Connected Vehicles

Quantum-Safe Security for the Connected Vehicle Ecosystem

<5ms

V2X Verification

1,500+

msg/sec Batch

72%

Smaller Certs

UNECE WP.29

Ready

Executive Summary

Quantum-safe V2X security for the connected vehicle ecosystem

<5ms

V2X Verification

1,500+

msg/sec Batch

72%

Smaller Certs

\$10-30

Per Vehicle/Year

54

WP.29 Countries

With 1.4 billion vehicles on roads globally and fleet lifecycles stretching 15-20 years into the future, vehicles sold today will still be operating in 2040-2045 — exactly when quantum computers are projected to break classical ECDSA cryptography. QBITEL Bridge delivers post-quantum V2X authentication, fleet-wide OTA PQC migration, and automated UNECE WP.29 compliance evidence in a single integrated platform.

1. The Connected Vehicle Security Crisis

Three threats that cannot be ignored

Threat	Description	Impact
V2X Spoofing	\$20 SDR device can inject false V2X messages, trigger emergency braking	Accidents, gridlock, platoon attacks
Quantum Fleet Lifecycle	15-20yr vehicle lifetime extends past quantum threat horizon (2040-2045)	Fleet-wide key compromise
UNECE WP.29 R155	Mandatory cybersecurity management in 54 countries — penalties include market withdrawal	Type approval revocation

2. V2X Message Authentication

IEEE 1609.2 + PQC — Cryptographic truth for every vehicle message

Attribute	Value
Algorithm	ML-KEM-768 (key encap) + ML-DSA-65 + Falcon-512 (signatures)
Compatibility	Backward compatible with existing RSU infrastructure
Coverage	BSM, SPaT, MAP, TIM, EVA message types
Latency	<5ms end-to-end signature verification

3. Implicit Certificate Compression

72% smaller than standard Dilithium — built for V2X bandwidth

Attribute	Value
Size	666 bytes (Falcon-512) vs 2,420 bytes (Dilithium)
Throughput	1,500+ msg/sec batch verification with SIMD
Encoding	X9.62 point compression + dictionary-based

Compatibility	IEEE 1609.2 extension field encoding
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4. Fleet-Wide OTA PQC Migration

Staged rollout with automatic rollback — zero fleet disruption

Attribute	Value
Rollout stages	Canary (10 vehicles) → 1% → 10% → 100%
Rollback	Automatic on >0.1% error rate at any stage
Delta updates	Cryptographic diff reduces OTA payload 80%
Verification	TPM-attested post-install verification

5. SCMS Integration & Pseudonym Management

Full Security Credential Management System compatibility

Attribute	Value
Pseudonyms	Short-lived certificates (1 week) for privacy
Misbehavior	Automated detection and reporting to SCMS
Revocation	CRL distribution in <100ms to all RSUs
Provisioning	Bulk enrollment: 10,000 vehicles/hour

6. Autonomous Vehicle Platooning Security

Authenticated convoy commands — rogue vehicles cannot join

Attribute	Value
Leader verification	Platoon leader PQC identity binding
Command auth	Every following distance/brake command signed
Rogue detection	<100ms detection of unauthorized convoy member
Failsafe	Graceful platoon disbanding on auth failure

7. ISO/SAE 21434 & UNECE WP.29 Compliance

Complete compliance evidence for type approval

Attribute	Value
TARA	Automated threat analysis and risk assessment
Evidence	Type approval security validation report
CSMS	Cybersecurity Management System documentation

R156	Software update management system compliance
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8. Compliance Coverage

All major automotive cybersecurity frameworks

Framework	Coverage	Key Requirement
UNECE WP.29 R155	Full	Cybersecurity Management System
UNECE WP.29 R156	Full	Software Update Management
ISO/SAE 21434	Full	TARA, security validation
IEEE 1609.2	Extended (PQC)	V2X security services
SAE J3061	Aligned	Cyber-physical systems guidebook
NIST FIPS 203/204	Native	ML-KEM, ML-DSA algorithms
SAE J2735	Compatible	DSRC/C-V2X message sets

9. Performance Specifications

Metric	Requirement	QBITEL Performance
V2X verification latency	<10ms	<5ms
Batch throughput	500+ msg/sec	1,500+ msg/sec
Certificate size	<2KB	666 bytes (Falcon-512)
OTA delivery time	<24 hours	<4 hours (delta)
False positive rate	<0.001%	<0.0001%
Fleet coverage stages	N/A	Canary → 1% → 10% → 100%
Cost per vehicle/year	\$50-200 (classical)	\$10-30 (QBITEL PQC)

10. Customer Scenarios

OEM

Global OEM — Fleet-Wide PQC Migration

- Challenge: 10M vehicle fleet, classical ECDSA expiring before 2040
- Solution: QBITEL staged OTA rollout with SCMS integration
- Result: UNECE WP.29 type approval, <\$30/vehicle/year cost
- Timeline: 12-month full fleet migration, zero downtime

TIER 1**Tier 1 Supplier — V2X Module Security**

- Challenge: Authentication chip integration for next-gen V2X modules
- Solution: Falcon-512 implicit certificates, SIMD batch verification
- Result: 1,500+ msg/sec, 72% smaller certificates vs Dilithium
- Timeline: 6-month silicon bring-up + validation

CITY**Smart City — V2I Infrastructure Security**

- Challenge: 500 RSUs with unauthenticated V2X broadcast
- Solution: RSU firmware update, traffic signal anti-spoofing
- Result: Cryptographic V2I authentication, intersection safety
- Timeline: 3-month RSU upgrade, zero road closure

11. Next Steps**STEP 1****V2X Security Assessment (Week 1-2)**

- Passive V2X traffic capture and protocol analysis
- Current SCMS architecture review
- UNECE WP.29 gap assessment

STEP 2**Proof of Concept (Month 1-2)**

- 1,000-vehicle pilot deployment
- SCMS integration and certificate provisioning
- Performance benchmark vs requirements

STEP 3**Fleet Migration Planning (Month 3)**

- Staged OTA rollout roadmap
- Rollback procedure documentation
- Type approval evidence package

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Email

Website

Schedule a call